

# DANG THO BACH

## Senior Java Engineer

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### PROFESSIONAL SUMMARY

- Back-end developer with almost 7 years of professional experience in Java and Spring Boot.
- Strong technical background with deep understanding of software development across Web (BE & FE) and Server technologies.
- Extensive experience in the Spring Ecosystem: Spring Boot, Spring Cloud, Spring Data JPA, Hibernate, Spring Security, Spring Batch, etc.
- Strong knowledge of relational database design, especially PostgreSQL and Oracle (PL/SQL optimization).
- Hands-on experience in CI/CD pipelines (GitLab, Jenkins) and containerization (Docker, Kubernetes).
- Adept at troubleshooting complex systems, collaborating with cross-functional teams, and delivering robust backend solutions.
- Experienced in integrating security tools and vulnerability remediation (SonarQube, Coverity SAST, Blackduck SCA).

### EDUCATION

#### Hanoi University Of Industry

Sep 2015 – June 2019

Specialization: Software Technology | GPA: 3.37 / 4.00

### PROFESSIONAL EXPERIENCE

#### VPBank (Vietnam Prosperity Joint-Stock Commercial Bank) — Senior Java Engineer

Apr 2025 – Present

#### Sai Gon – Ha Noi Commercial Joint Stock Bank (SHB) — Software Engineer

Sep 2022 – Mar 2025

#### Alphaway Technology Company — Software Engineer

Aug 2019 – Aug 2022

#### Ominext Joint Stock Company (JSC) — Java Developer

Dec 2018 – Jun 2019

### KEY PROJECTS

## PDMS Service — Physical Document Management System

Apr 2025 – Present (VPBank)

### Role: Senior Java Engineer (Project entering maintenance/CR phase)

Java 21 · Spring Boot 3.4 · PostgreSQL · Apache Kafka · Keycloak · AWS S3 · Redis · Docker · ShedLock · Resilience4j

Led architecture design and development of the core backend service for VPBank's enterprise Physical Document Management System — a bank-wide platform managing the full lifecycle of 7M+ physical banking documents across 50+ warehouse locations, serving more than 1000 internal users daily. Delivered 20+ release candidates in 14 months, transitioning the project from greenfield to production stability.

### SCALE & IMPACT

- **Domain & API Scope:** Engineered 239+ REST API endpoints across 53 components, organized into 7 functional domains (accounting records, credit documents, distributed documents, warehouse operations, reporting, system admin, and RBAC).
- **Large-Scale Data Migration:** Migrated 7M+ legacy credit case records (CasePDM) from multiple source systems via multi-phase PostgreSQL stored procedures, processing 5,000–10,000 records per batch with session-scoped lookup tables, composite indexes, and FOR UPDATE SKIP LOCKED to prevent concurrent deadlocks.
- **Incremental Ingestion:** Designed and optimized a weekly incremental data ingestion pipeline from upstream systems (Jarvis, EDA Card, SFTP sources) to handle sustained throughput under high cumulative entity growth.

### AI-AUGMENTED SDLC (DEV CYCLE REDUCED BY ~40%)

- **Requirements Analysis:** Utilized Kiro + Confluence MCP integration to auto-read Confluence specs, perform gap analysis between requirements and existing code, and generate structured technical documents.
- **Sprint Planning Optimization:** Integrated Kiro + Jira MCP to auto-generate subtasks, acceptance criteria, and implementation plans from Jira epics/stories, reducing planning overhead by ~60%.
- **Code Generation & Review:** Leveraged AI-assisted coding for boilerplate patterns (CRUD endpoints, DTO mappings, validation logic, migration scripts), with human review focusing on core business logic (achieving 20+ RCs in 14 months).
- **Test Generation:** Automated unit test scaffolding for 150+ test classes using property-based testing (jqwik), maintaining 80%+ JaCoCo coverage.
- **Documentation Sync & RAG:** Automated API spec review (Swagger auditing) and Confluence updates. Configured an internal RAG MCP server backed by Bedrock for semantic code search and domain knowledge retrieval.
- **Test Management:** Integrated Zephyr MCP to automate test case creation and execution tracking directly mapped to Jira stories.

### LARGE-SCALE DATA PROCESSING SOLUTIONS

- **Streaming Excel Export:** Built a custom SXSSFStreamingWriter framework on Apache POI SXSSF to support memory-efficient Excel exports of 100K+ rows with constant heap usage (paginated reads, lazy initialization).
- **Async Reporting:** Implemented asynchronous report generation (@Async + dedicated thread pools) with automated AWS S3 upload, preventing blocking of application request threads.
- **Configurable SFTP Pipeline:** Designed a chunk-based streaming pipeline (CSV/Excel) with a dynamic SQL rule engine, enabling zero-code onboarding of new data sources with configurable batch sizes.
- **Batch Database Tuning:** Optimized stored procedures using session tables with targeted indexes, CTE chains for parallel business rule validation (10+ rules per record), and chunked commits to prevent WAL bloat.

### ARCHITECTURE & ENGINEERING LEADERSHIP

- **Clean Architecture:** Enforced Hexagonal Architecture (ports & adapters) to separate core domain, application, and infrastructure layers, facilitating rigorous testing.
- **Event-Driven Integration:** Deployed Apache Kafka (SASL\_SSL, SCRAM-SHA-512) for SFTP notifications, BPM workflow callbacks, and authorization snapshot events.
- **Distributed Coordination & Security:** Used ShedLock over JDBC to prevent duplicate execution of 15+ scheduled jobs. Integrated Keycloak OAuth2/OIDC with custom Gateway filters to enforce service-layer RBAC. Patched 15+ high/critical CVEs.

### DATA MIGRATION & INTEGRITY

- **Remediation Scripts:** Identified 5 critical business gaps and resolved structural data integrity issues where credit documents were incorrectly linked across multiple warehouses; implemented idempotent 10-step cleanup scripts.
- **Maker-Checker Workflow:** Designed batch confirmation workflows with pessimistic locking, maker-checker validation, and post-commit notifications, supporting 500 documents per request with transactional safety.

## Corebank Extension & Hygiene System

Sep 2022 – Mar 2025 (SHB)

**Role: Software Engineer (Corebank Intellect) | Team Size: 10**

Spring Boot · REST API · Oracle (PL/SQL) · Spring Data JPA · Docker · Kubernetes · BlackDuck · Coverity

Developed transactional modules and compliance extensions surrounding SHB's core-banking platform.

- Maintained and fixed issues for the web system on demand.
- Developed and integrated the email notification system for lending and tide modules.
- Developed and optimized batch jobs to process update of expired identity documents in compliance with Circular 17/2024/TT-NHNN of the State Bank of Vietnam, processing millions of records in seconds.
- Integrated BlackDuck (SCA) and Coverity (SAST) security pipelines to perform dependency and static code scanning prior to deployment.

## Corecard Integration & Migration

Sep 2022 – Mar 2025 (SHB)

**Role: Software Engineer (Corecard SmartVista) | Team Size: 4**

Spring Boot · REST API · Oracle (PL/SQL) · Spring Data JPA · Keycloak IDP · Docker · Kubernetes · GitLab CI/CD · ELK Stack · JasperReports

Spearheaded the migration of the SmartVista core card system utility (CardUtility) and deployed enterprise identity management.

- Deployed Keycloak Identity Provider (IDP) and configured Single Sign-On (SSO) integration across the organization.
- Migrated 17 core CardUtility functions (card/PIN issuance, Visa/Mastercard reporting, ATM/POS management, smartvisa CBS reporting) to meet upgraded SVNG core requirements.
- Redesigned CardUtility into microservices containerized on Kubernetes and established GitLab CI/CD automation pipelines.
- Deployed ELK stack on containers for centralized resource monitoring and log tracing.
- Developed high-performance batch updates for expired identity documents and built extensive reports using JasperReports.

## Way4Batch Scheduling System

Jan 2022 – Aug 2022 (Military Bank)

**Role: Software Engineer | Team Size: 13**

Spring Boot · SOAP/REST WebServices · JDBC · Oracle (PL/SQL) · Multithreading

Built a high-concurrency job scheduling system handling card lot processing and transactional banking tasks.

- Built multithreaded job scheduling for card issuance, payment accounting, card reissuing, and limit changes.
- Managed and completed JDK upgrade from 1.6 to 11 with zero service disruption.
- Collaborated in an Agile/Scrum team to deliver request improvements and meet sprint goals.

## Card Issue Gateway

Jan 2022 – Aug 2022 (Military Bank)

**Role: Software Engineer | Team Size: 13**

Spring Boot · SOAP/REST WebServices · JDBC · Oracle (PL/SQL)

Developed gateway services for partner integrations.

- Built up gateway APIs for issuing UnionPay card and virtual card (card-by-code).
- Completed sprint goals under Scrum model.

## CardPortal System

Jan 2022 – Aug 2022 (Military Bank)

**Role: Software Engineer | Team Size: 13**

Windows Server · ExtJS · SOAP WebServices · Oracle (PL/SQL) · Way4 Corecard

Built services and management screens for card operation workflows.

- Developed functionalities for card issuing, report generation, PIN management, card reissuing, customer updates, and stoplist/blacklist card.

## CardGateway Notification System

Jan 2022 – Aug 2022 (Military Bank)

**Role: Software Engineer | Team Size: 3**

Windows Server • ExtJS • SOAP WebServices • Oracle (PL/SQL) • Way4 Corecard

Card management system for operation and card lot processing counter channels.

- Developed partner channel configuration screens and set up cryptographic keys.
- Configured SMS message template and distribution for balance changes, and active/close MCredit/MB card events.

## Wealth Management System

Oct 2021 – Dec 2021 (Military Bank)

**Role: Software Engineer | Team Size: 30**

Spring Cloud • Keycloak • SOAP/REST WebServices • Oracle

Built core modules for an investment and wealth management system.

- Configured user administration and access control using Keycloak.
- Implemented portfolio restructuring and financial settlement algorithms for wealth experts.
- Built up customer profile CRUD APIs.

**Wallet Core Conversion (ViettelPay)**

Jan 2021 – Sep 2021 (Military Bank)

**Role: Software Engineer | Team Size: 7**

Spring Boot · Oracle · SOAP/REST WebServices · Socket ISO8583

Migrated the ViettelPay wallet connection layer off the core card system.

- Modified and constructed socket connection streams (ISO8583) to wallet core for: balance queries, card locking/unlocking, beneficiary lookup, customer status synchronization, and transaction history.
- Developed app-based cash withdrawal feature (Cash-by-code).

**Credit Rating System**

Jan 2021 – Jun 2021 (An Binh Bank)

**Role: Software Engineer | Team Size: 4**

iDempiere Server · ZK Framework · JDBC · Oracle

Developed an enterprise credit rating system for individuals and corporate businesses.

- Built interactive credit scoring interfaces, customer record management screens, and Credit Information Center (CIC) records integration.
- Developed score rating templates and export functionality via JasperReports.

**Credit Rating System**

Apr 2020 – Dec 2020 (Lien Viet Postbank)

**Role: Software Engineer | Team Size: 7**

iDempiere Server · ZK Framework · JDBC · Oracle

Developed an enterprise credit rating system for individuals and corporate businesses.

- Built scoring screens, customer records, and CIC records matching workflows.
- Built scoring templates and ratings reporting on JasperReports.

**Government eCabinet meeting portal**

Aug 2019 – Mar 2020 (Government Client)

**Role: Software Engineer | Team Size: 8**

J2EE · JSP / Servlet · JDBC · DB2 · IBM WebSphere

Designed and built modules for the official government cabinet meeting and voting system.

- Developed screens for news feed, meeting coordination, real-time voting, and role-based permissions.

**Secom Electronic Medical Record System**

Mar 2019 – June 2019 (Japanese Client)

**Role: Java Developer | Team Size: 7**

J2EE · JSP / Servlet · JDBC · DB2 · IBM WebSphere

Maintained electronic medical record systems and resolved on-demand system issues.

**Fine Medical Pharmacy System**

Dec 2018 – Apr 2019 (Japanese Client)

**Role: Java Developer | Team Size: 6**

SpringBoot · Hibernate · JSF · MySQL

Developed pharmacy management and inventory control CRUD interfaces.